PORTFOLIO PROJECT BRIEF

Insurance Charges Prediction

Multiple Linear Regression | Insurance / Financial Services

Problem Description

Health insurers and healthcare payers price policies and negotiate reimbursement rates by estimating how much a given claim or procedure is likely to cost. Real billing data is rarely as tidy as a textbook dataset — costs are stored as currency strings, categories are inconsistent, and there are far more distinct procedure and provider categories than in a toy example. Underwriting, actuarial, and healthcare-finance analysts have to clean and model this kind of data as a matter of routine.

In this project, you will play the role of a data scientist working with a healthcare payer's pricing/actuarial team. You will use **real U.S. government hospital billing data** to build a multiple linear regression model that predicts the **average amount a hospital charges** for a given inpatient procedure, based on the procedure type, location, and volume of discharges. You will then interpret the model's coefficients to produce a pricing/cost-variation insight that a non-technical stakeholder (e.g. a payer contracting manager) could act on.

Dataset

Recommended dataset: Hospital Charges for Inpatients, sourced from real, publicly released U.S. Centers for Medicare & Medicaid Services (CMS) data, available on Kaggle: **kaggle.com/datasets/speedoheck/inpatient-hospital-charges**. It contains roughly 163,000 records (one row per hospital per procedure type) with the following columns:

Column	Description
DRG Definition	The diagnosis-related group: the billed procedure/condition category (~100 distinct values)
Provider Id / Provider Name	Identifier and name of the hospital (high-cardinality — not useful as a raw model feature)
Provider Street Address, City, Zip Code	Hospital location detail (very high-cardinality — drop for modelling)
Provider State	US state the hospital is located in
Hospital Referral Region Description	Regional healthcare market the hospital belongs to
Total Discharges	Number of Medicare patient discharges billed under this DRG at this hospital
Average Covered Charges	What the hospital billed, on average, for this DRG (target variable)
Average Total Payments	What was actually paid on average, including patient contributions
Average Medicare Payments	What Medicare itself paid on average

Why this is 'real world': the three charge/payment columns are stored as text with a leading '$' sign (e.g. *"$32,963.07"*) rather than as clean numbers, *DRG Definition* has around 100 distinct categories rather than 2-4, and several columns (exact address, provider name) are too granular to use directly and need a conscious decision to exclude. This is exactly the kind of light but genuine cleaning you'll encounter with real billing or claims data — just at a manageable scale (one CSV, ~163k rows, no joins required).

If this exact dataset is unavailable, any real (not pre-cleaned-for-tutorials) claims/billing dataset with a currency-formatted target and at least one categorical column with dozens of categories will work equally well for this brief.

Learning Objectives

- Practise cleaning real-world 'dirty' data: currency strings, whitespace in column names, and columns that must be deliberately excluded because they are identifiers rather than predictors.
- Practise encoding a categorical variable with many categories (DRG Definition), not just 2-4.
- Practise end-to-end multiple linear regression: EDA, encoding, train/test split, fitting, evaluation.
- Practise engineering a derived business metric (a cost ratio) from existing columns.
- Practise applying and comparing regularised regression (Ridge/Lasso) against plain OLS.
- Practise interpreting regression coefficients in a real business (healthcare pricing) context.
- Practise evaluating a regression model using RMSE and R^2, and visualising errors with a residual plot.

Task Objectives & Deliverables

Create a Jupyter notebook named **hospital_charges_prediction.ipynb** and complete the following steps:

1. **Load the data.** Import the CSV into a pandas DataFrame. Inspect its shape, dtypes, and column names (note that column names may contain leading/trailing spaces that need stripping).
2. **Clean the currency columns.** *Average Covered Charges*, *Average Total Payments*, and *Average Medicare Payments* are stored as text (e.g. *"$32,963.07"*). Strip the '$' sign and comma separators and convert them to numeric (float) columns.
3. **Reduce cardinality.** Drop columns that are identifiers rather than useful predictors for a general pricing model: *Provider Id*, *Provider Name*, *Provider Street Address*, and *Provider Zip Code*. Keep *Provider State* and *DRG Definition* as your categorical predictors.
4. **Exploratory data analysis.** Plot the distribution of *Average Covered Charges*. Show, using a bar chart or box plot, how average charges vary across the 10-15 most frequent DRG categories, and across states. Note which variable appears to explain more of the variation.
5. **Handle missing values and duplicates.** Check for missing values and duplicate rows and handle them appropriately, documenting your reasoning in a markdown cell.

6. **Encode categorical variables.** One-hot encode *Provider State*. For *DRG Definition* (~100 categories), either one-hot encode it directly, or first group any DRGs that appear fewer than a chosen number of times into an 'Other' category to keep the number of columns manageable — explain your choice in a markdown cell.
7. **Define X and y.** Assign your independent variables to **X** (use *Total Discharges* plus your encoded categorical columns; do not include *Average Total Payments* or *Average Medicare Payments*, since these would leak information about the target). Assign *Average Covered Charges* to **y**.
8. **Train/test split.** Split the data into training and test sets using an 80:20 ratio.
9. **Fit the model.** Use *sklearn.linear_model.LinearRegression* to fit a multiple linear regression model on the training set.
10. **Print model parameters.** Print the intercept and the coefficient for each feature.
11. **Generate predictions.** Use the trained model to predict charges for the test set.
12. **Evaluate the model.** Compute RMSE and R^2 on the test set using *sklearn.metrics*. Comment on what the R^2 value tells you about how well procedure type, state, and discharge volume explain billed charges.
13. **Residual plot.** Plot predicted values against residuals (actual minus predicted) to visually check for systematic errors — e.g. does the model consistently under-predict for high-cost procedures?
14. **Interpret coefficients.** In a markdown cell, identify the DRG category and/or state associated with the largest positive coefficients, and explain in plain language what this tells a payer contracting manager about where cost variation is coming from.
15. **Engineer a business metric.** Create a *charge-to-payment ratio* column (Average Covered Charges ÷ Average Medicare Payments) and identify the 5 DRGs and/or states with the largest average mark-ups. This is the kind of derived metric a pricing/actuarial analyst is regularly asked to produce.
16. **Compare against an alternate target.** Refit your model (same features, same train/test split) to predict *Average Medicare Payments* instead of *Average Covered Charges*. Compare the RMSE, R^2, and coefficients between the two models, and comment on what differs between what hospitals bill and what Medicare actually pays.
17. **Compare against regularised regression.** Refit your original model using *Ridge* and *Lasso* from *sklearn.linear_model* (these are especially relevant once you have many one-hot encoded DRG/state columns). Compare RMSE and R^2 on the test set against your plain OLS model, and comment on whether regularisation changed which features had the largest coefficients.
18. **Summarise findings.** Write a short summary (5-8 sentences) of your findings — including the regularisation and alternate-target comparisons — and one recommendation you would make to a healthcare payer based on your model.